

7.8 Quiz: Logarithms (no calculator)

1. [Maximum mark: 5]

Write each of the following expressions in the form $\ln k$, where $k \in \mathbb{Z}^+$.

(a) $\ln 4 + \ln 7$ [1]

(b) $2 \ln 5$ [2]

(c) $-\ln \frac{1}{6}$ [2]

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2. [Maximum mark: 5]

Let $\log_{10} 3 = p$ and $\log_{10} 5 = q$.

(a) Find an expression for $\log_{10} 45$ in terms of p and q . [3]

(b) Find an expression for $\log_3 25$ in terms of p and q . [2]

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3. [Maximum mark: 5]

It is given that $\log_{10} b = \frac{2}{3}$, where $b > 0$.

Find the value of

$$\text{(a) } \log_{10}\left(\frac{1}{b}\right); \quad [2]$$

(b) $\log_{1000} b$. [3]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is a vertical margin line on the left side, creating a narrow left margin. The paper appears to be from a notebook or a standard ruled document.

4. [Maximum mark: 5]

Solve the equation $2 \ln x = \ln 16 + 4$. Give your answer in the form $x = pe^q$ where $p, q \in \mathbb{Z}^+$.